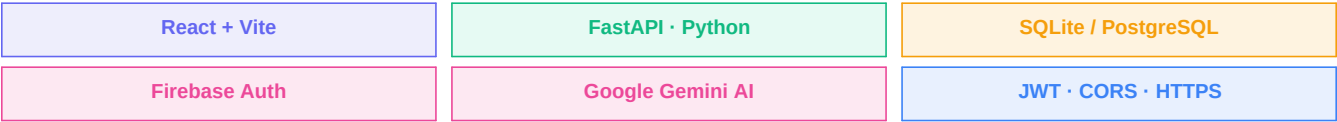


■ DataViz — Architecture Scheme

Full-stack SaaS data visualization platform · React + FastAPI · v3.0.0



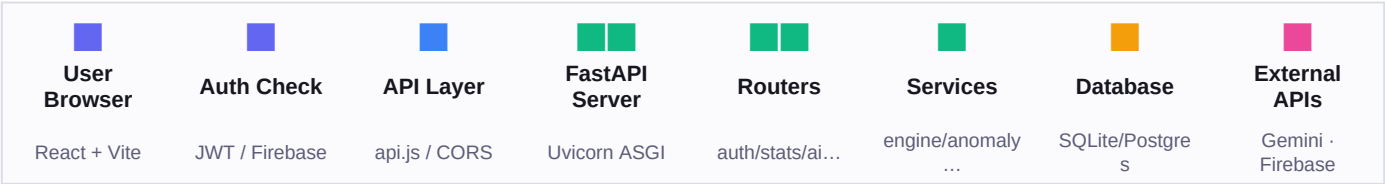
- Frontend (React)
- Backend (FastAPI)
- Database / Storage
- External Services
- Infrastructure / Security

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1. Request Data Flow

A single user request travels through the following stages, from the browser down to persistent storage and back out through any external AI/auth providers involved:



Flow order: Browser → Auth Check → API Layer → FastAPI Server → Routers → Services → Database, with External APIs (Gemini, Firebase) invoked on demand by the services layer.

2. System Overview

Frontend — React + Vite

■ Auth Layer	auth/
AuthContext, ProtectedRoute, firebase.js · JWT token handling	
■ Context Providers	context/
DataContext (dataset state) · ThemeContext (dark/light)	
■ API Client	api/
Axios instance · api.js, contact.js, gemini.js	
■ Custom Hooks	hooks/
useAIAnalysis · useSmartCleaning	
■ Client Services	services/
aiAdvisorService · ruleEngine (data-cleaning rules)	

Backend — FastAPI v3.0

■ App Core (app/)	app/
models, database, user_store, data_context, dataset_manager, cache	
■ Routers (10 modules)	routers/
auth · admin · stats · analyze · charts · ai_advisor · firebase_auth · gemini · profiling · contact	
■ Services (7 modules)	services/
engine · anomaly · quality_score · stats_service · chart_service · analyzer_service · main	
■ Utils	utils/
security (bcrypt, JWT) · file_utils	
■ Core Config	core/
config · firebase_config · supabase_client	

External Services

■ Firebase Auth

Google OAuth · email/password · firebase-admin SDK

ext

■ Google Gemini AI

AI chat assistant · AI advisor · data insights generation

ext

■ Supabase (optional)

Remote PostgreSQL backend · file storage

ext

■ Redis Cache

In-memory dataset cache · session TTL

ext

3. Frontend Pages & Routes

PUBLIC / — Home Page	PUBLIC /about — About
PUBLIC /login — Auth (Login / Register)	PUBLIC /forgot-password — Password recovery
PUBLIC /features — Features	PUBLIC /pricing — Pricing
PUBLIC /faq, /contact — FAQ & Contact	PUBLIC /privacy, /terms, /cookies — Legal pages
AUTH /upload — File Upload	AUTH /dashboard — Main Dashboard
AUTH /analytics — Analytics	AUTH /stats — CSV Stats
AUTH /data-quality — Data Quality	AUTH /3d-visualizations — 3D Charts
AUTH /model — ML Model Builder	AUTH /user — Profile
ADMIN /admin — Admin Panel	404 * — Not Found

4. Key Frontend Components

■ AIAdvisor

- AIChatAssistant
- AIAdvisorPanel
- IssueCard
- MLReadinessScore
- SmartFixButton

■ DataQuality

- ColumnProfiler
- QualityMetrics
- RecommendationEngine

■ Charts & Viz

- ChartBuilder
- ChartContainer
- MatplotlibChart
- HeatmapDisplay

■ Data Tools

- CleaningToolbar
- AutoCleanAgent
- ExcelGrid
- FileUploader

■ Stats & Models

- StatsPanel
- CsvStatsTable
- DescriptiveStatsTable
- ModelBuilderSection
- PredictionSection

5. Backend API Endpoints

POST /api/auth/login	POST /api/auth/register
POST /api/auth/refresh	POST /api/auth/firebase
POST /api/analyze/upload	GET /api/analyze/info
GET /api/analyze/preview	POST /api/analyze/reset
GET /api/stats/describe	GET /api/stats/correlation
GET /api/stats/distribution	GET /api/stats/outliers
POST /api/charts/build	POST /api/charts/matplotlib
POST /api/ai-advisor/analyze	POST /api/ai-advisor/chat
POST /api/ai-advisor/fix	POST /api/gemini/chat
GET /api/profiling/report	GET /api/admin/users
DEL /api/admin/users/{id}	GET /api/admin/metrics
GET /api/health	POST /api/contact

6. Database Schema

■ users

id	UUID PK
firebase_uid	VARCHAR?
name	VARCHAR
email	VARCHAR UNIQUE
password_hash	VARCHAR?
role	user/admin
is_active	BOOL
uploads	INT
api_calls_count	INT
last_login	DATETIME

■ datasets

id	UUID PK
user_id	FK→users
filename	VARCHAR
file_path	TEXT
rows	INT
columns	INT
file_size	BIGINT
created_at	DATETIME

■ sessions

id	UUID PK
user_id	FK→users
token	TEXT
expires_at	DATETIME
is_revoked	BOOL
created_at	DATETIME

■ activity_logs

id	INT PK
user_id	FK→users
action	VARCHAR
resource	VARCHAR
details	JSON
ip_address	VARCHAR
created_at	DATETIME

7. Backend Services Layer

■ Analysis Engine

■ engine.py

Core data processing pipeline · Pandas/Polars

■ analyzer_service.py

Column type inference · data profiling

■ stats_service.py

Descriptive stats · correlation · distribution

■ chart_service.py

Plotly / Matplotlib chart generation

■ Quality & AI

■ quality_score.py

Data quality scoring · completeness · consistency

■ anomaly.py

Anomaly detection · outlier flagging

■ ai_advisor.py

Gemini-powered analysis · chat · auto-fix

■ stats.py

87 KB · Comprehensive statistics module

■ Auth & Admin

■ routers/auth.py

Register / login / refresh · bcrypt + PyJWT

■ firebase_auth.py

Firebase token verify · user sync

■■ admin.py

User management · metrics · activity logs

■ utils/security.py

Password hashing · JWT encode/decode

8. Infrastructure & Security

■ CORS Middleware

■ HTTPS Redirect (prod)

■■ XSS / CSRF Headers

■■ Request Logging

■ Global Exception Handler

■ JWT Auth (PyJWT)

■ bcrypt Password Hashing

■ Firebase Admin SDK

■ Redis In-Memory Cache

■ SQLAlchemy ORM

■ PostgreSQL (prod)

■■ SQLite (dev)

■ Pagination Headers

■ OpenAPI / Swagger Docs

■ Uvicorn ASGI

■ dotenv Config

9. Technology Stack

■ ■ Frontend Stack

■ ■ React 18

■ Vite

■ TailwindCSS

■ React Router v6

■ Axios

■ Firebase SDK

■ Plotly.js

■ Dark/Light Theme

■ Backend Stack

■ FastAPI

■ Uvicorn

■ Pandas

■ NumPy

■ Polars

■ Plotly

■ Matplotlib

■ SciPy

■ scikit-learn

■ rapidfuzz

■ ■ External / AI

■ Google Gemini AI

■ Firebase Auth

■ Supabase

■ Redis

■ PostgreSQL